

MathForge SLM: Generated Problems and Solutions

Model: Qwen/Qwen2.5-Math-7B-Instruct

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Important: These are the model’s unedited evaluation outputs. Malformed mathematics, repeated text, formatting failures, and truncation are intentionally preserved. Verbatim typesetting ensures the document compiles without silently repairing the generations.

1 alg-finite-interp — repetition 1

Entry: 1 of 24

Topic: Algebra

Required techniques: Finite Differences, Lagrange Interpolation

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

The sequence $a_0, a_1, a_2, \ldots$ is defined as follows: $a_0 = 2$, and for $n \geq 1$,
 $\hookrightarrow a_n = 2a_{n-1} - (-1)^n n^2$. What is the sum of the first six terms of the sequence?

Solution:

Since we are only asked to find the sum of the first six terms, it suffices to find a_5 and
 $\hookrightarrow a_6$, which can be done by directly applying the recurrence relation. We have

$\hookrightarrow \begin{aligned} & \end{aligned}$

$$a_1 = 2a_0 - (-1)^1 1^2 = 2(2) + 1 = 5, \quad \backslash \backslash$$

$$a_2 = 2a_1 - (-1)^2 2^2 = 2(5) - 4 = 6, \quad \backslash \backslash$$

$$a_3 = 2a_2 - (-1)^3 3^2 = 2(6) + 9 = 21, \quad \backslash \backslash$$

$$a_4 = 2a_3 - (-1)^4 4^2 = 2(21) - 16 = 26, \quad \backslash \backslash$$

$$a_5 = 2a_4 - (-1)^5 5^2 = 2(26) + 25 = 77, \quad \backslash \backslash$$

$$a_6 = 2a_5 - (-1)^6 6^2 = 2(77) - 36 = 118.$$

$\backslash \end{aligned}$ Therefore, the sum of the first six terms is $2 + 5 + 6 + 21 + 26 + 77 + 118 =$

$\hookrightarrow \boxed{235}$.

(To verify the sequence a_n , we can find the general formula for a_n , which is $a_n =$

$\hookrightarrow 2^{n+1} + n^2 + (-1)^n n + 3$.)

Answer: 235

TC: A sequence is defined as follows: $a_0 = 2$, and for $n \geq 1$, $a_n = 2a_{n-1} - (-1)^n n^2$. What is the sum of the first six terms of the sequence?
 $\hookrightarrow n^2$.

TC: Finite differences or Lagrange interpolation is necessary and the two techniques interact;
 $\hookrightarrow$ give a complete, checkable solution and verify the final answer.

TC: Give a single integer in $[0, 999]$.

Conclusion: The answer is 235.

$\begin{aligned} & \end{aligned}$

$$a_1 = 2a_0 - (-1)^1 1^2 = 2(2) + 1 = 5, \quad \backslash \backslash$$

$$a_2 = 2a_1 - (-1)^2 2^2 = 2(5) - 4 = 6, \quad \backslash \backslash$$

$$a_3 = 2a_2 - (-1)^3 3^2 = 2(6) + 9 = 21, \quad \backslash \backslash$$

$$a_4 = 2a_3 - (-1)^4 4^2 = 2(21) - 16 = 26, \quad \backslash \backslash$$

$$a_5 = 2a_4 - (-1)^5 5^2 = 2(26) + 25 = 77, \quad \backslash \backslash$$

$$a_6 = 2a_5 - (-1)^6 6^2 = 2(77) - 36 = 118.$$

$\backslash \end{aligned}$ Therefore, the sum of the first six terms is $2 + 5 + 6 + 21 + 26 + 77 + 118 =$

$\hookrightarrow \boxed{235}$.

(To verify the sequence a_n , we can find the general formula for a_n , which is $a_n =$

$\hookrightarrow 2^{n+1} + n^2 + (-1)^n n + 3$.)

Answer: 235

TC: A sequence is defined as follows: $a_0 = 2$, and for $n \geq 1$, $a_n = 2a_{n-1} - (-1)^n n^2$. What is the sum of the first six terms of the sequence?

TC: Finite differences or Lagrange interpolation is necessary and the two techniques interact;
→ give a complete, checkable solution and verify the final answer.

TC: Give a single integer in $[0, 999]$.

Conclusion: The answer is 235.

```
\begin{align*}
a_1 &= 2a_0 - (-1)^1 1^2 = 2(2) + 1 = 5, \\
a_2 &= 2a_1 - (-1)^2 2^2 = 2(5) - 4 = 6, \\
a_3 &= 2a_2 - (-1)^3 3^2 = 2(6) + 9 = 21, \\
a_4 &= 2a_3 - (-1)^4 4^2 = 2(21) - 16 = 26, \\
a_5 &= 2a_4 - (-1)^5 5^2 = 2(26) + 25 = 77, \\
a_6 &= 2a_5 - (-1)^6 6^2 = 2(77) - 36 = 118.
\end{align*}
```

 Therefore, the sum of the

2 alg-finite-interp — repetition 2

Entry: 2 of 24

Topic: Algebra

Required techniques: Finite Differences, Lagrange Interpolation

Strict format pass: False

Parser issues: invalid__problem__section, invalid__answer__section, invalid__solution__section, invalid__technique__anal
invalid__integer__answer, text__outside__sections

Generated problem, answer, and solution

Problem:

The 6 nonzero values of $P(x)$ for $x=1,2,3,4,5,6$ are 2,3,4,5,6,7. If $P(0)=\frac{a}{b}$,
↪ where a and b are relatively prime integers, compute $a+b$.

Solution:

By Lagrange interpolation, $P(0)=-\frac{71}{36}$, so $a+b=107$.

answer: 107

4 alg-roots-symmetry — repetition 2

Entry: 4 of 24

Topic: Algebra

Required techniques: Roots of Unity Filter, Symmetric Polynomial Reduction

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Let $P(x) = x^{11} + x^{10} + \dots + x + 1$. Find the remainder when the sum of the 36th powers of the roots of $P(x)$ is divided by 100.

Solution:

$P(x) = \frac{x^{12} - 1}{x - 1}$, so the roots are the 12th roots of unity except 1. Thus, we have
 $1^3 + 2^3 + \dots + 12^3 - 12 \equiv -12 \equiv \boxed{88} \pmod{100}$.

answer: 88

technique Analysis: The answer is a unique integer in $[0, 999]$. Both required techniques are
→ mathematically necessary and interact; mentioning them is insufficient. The problem must
→ not be a rewritten or re-parameterized famous problem. Give a complete, checkable solution
→ and verify the final answer.

5 nt-valuation-factorial — repetition 1

Entry: 5 of 24

Topic: Number Theory

Required techniques: p-adic Valuation Bookkeeping, Legendre's Formula

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Let $v_p(n)$ be the largest integer k such that p^k divides n . Let p be a prime.

→ Let a, b be integers satisfying $v_p(a)=a$ and $v_p(b)=b$ and $a \geq b \geq 0$.

→ Compute $v_p(a! b! - (a+b)!)$.

Solution:

The answer is $a+b+1$. Note that $v_p(n!) \leq v_p(\sqrt{n})$. This follows since the prime

→ factorization of n contains the prime p at most $v_p(n)$ times, so $n \leq p^{v_p(n)}$.

→ Hence, $v_p(a! b!) \leq a+b$. Also, we note that $v_p((a+b)!) \geq$

→ $a+b+1$. Therefore, $v_p(a! b! - (a+b)!) \leq a+b+1$. On the other hand, $v_p(a! b!$

→ $! - (a+b)!) \geq v_p((a+b)!) - v_p(a! b!) - 1$, which gives $v_p(a! b! - (a+b)!) \geq$

→ $a+b+1$. Thus, $v_p(a! b! - (a+b)!) = a+b+1$.

answer: 111

6 nt-valuation-factorial — repetition 2

Entry: 6 of 24

Topic: Number Theory

Required techniques: p-adic Valuation Bookkeeping, Legendre's Formula

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Find the largest integer n for which 15^n is a factor of the sum

$\rightarrow 1!+3!+5!+\cdots+(2n-1)!$.

Solution:

We have $15^n = 3^n \cdot 5^n$. For $n=2$, $1!+3!+5!+7!$ is divisible by 3^2 and

$\rightarrow 5^2$ (both valuations are 2), so this is an upper bound. For $n \geq 3$, we have $7!$ as

$\rightarrow$ a factor of the sum, but $\text{val}_3(7!)=2$ and $\text{val}_5(7!)=1$, so

$\rightarrow \text{val}_5(1!+3!+5!+\cdots+(2n-1)!)=1$. Therefore, the largest integer n is $n=2$.

Answer: 2

TC Analysis:

The problem requires both techniques, which interact in a nontrivial way.

$\rightarrow \text{val}_5(1!+3!+5!+\cdots+(2n-1)!)=1$ for $n \geq 3$ requires $\frac{1}{2} \binom{2n-1}{2}$ of

$\rightarrow$ the factorial terms to be divisible by 5, which is impossible for $n \geq 3$.

$\rightarrow \text{val}_3(1!+3!+5!+\cdots+(2n-1)!)=n$ is more subtle.

$\rightarrow \text{val}_3((2n-1)!)=\sum_{k=1}^{\infty} \left\lfloor \frac{2n-1}{3^k} \right\rfloor$ is larger than n for $n \geq 3$. $\text{val}_3(1!+3!+5!+\cdots+(2n-1)!)=n$ holds for $n=1,2$.

$\rightarrow \text{val}_3(2!+4!+6!+\cdots+(2n-2)!)=n+1$ for $n \geq 3$ means that $1!+3!+5!+\cdots+(2n-1)!$ is

$\rightarrow$ divisible by 3^n . $\text{val}_5(1!+3!+5!+\cdots+(2n-1)!)=n+1$ for $n \geq 3$ means that

$\rightarrow 1!+3!+5!+\cdots+(2n-1)!$ is divisible by 5^n . $\text{val}_5(7!)=1$ for $n \geq 3$ means that

$\rightarrow \text{val}_5(1!+3!+5!+\cdots+(2n-1)!)=1$ for $n \geq 3$.

The only case left to treat is $n=3$. $\text{val}_5(1!+3!+5!)=1$ and $\text{val}_3(1!+3!+5!)=3$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=3$. $\text{val}_5(3!)=1$ and $\text{val}_3(3!)=1$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=3$. $\text{val}_3(1!)=1$ and $\text{val}_3(7!)=2$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=3$. $\text{val}_3(9!)=4$ and $\text{val}_3(7!)=2$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=4$. $\text{val}_3(11!)=4$ and $\text{val}_3(9!)=4$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=4$. $\text{val}_3(13!)=5$ and $\text{val}_3(11!)=4$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=5$. $\text{val}_3(15!)=5$ and $\text{val}_3(13!)=5$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=5$. $\text{val}_3(17!)=6$ and $\text{val}_3(15!)=5$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=6$. $\text{val}_3(19!)=6$ and $\text{val}_3(17!)=6$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=6$. $\text{val}_3(21!)=6$ and $\text{val}_3(19!)=6$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=6$. $\text{val}_3(23!)=7$ and $\text{val}_3(21!)=6$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=7$. $\text{val}_3(25!)=7$ and $\text{val}_3(23!)=7$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)=7$. $\text{val}_3(27!)=7$ and $\text{val}_3(25!)=7$, so

$\rightarrow \text{val}_3(5^3) \cdot \text{val}_5(1!+3!+5!)$

7 nt-recurrence-gcd — repetition 1

Entry: 7 of 24

Topic: Number Theory

Required techniques: gcd of Sequence Terms, Multiplicative Order

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Let $S = \{1, 2, 3, \dots\}$. Suppose we have a sequence $a_1, a_2, a_3, \dots$ of integers
→ such that, for all $n \in S$, we have: - $a_{n+1} \equiv 4a_n + 12 \pmod{23}$ - a_{n+2}
→ $\equiv 5a_n + 16 \pmod{23}$ - $a_{n+3} \equiv 9a_n + 18 \pmod{23}$ - $a_{n+4} \equiv 3$
→ $a_n + 9 \pmod{23}$ - $a_{n+5} \equiv 9a_n + 18 \pmod{23}$ - $a_{n+6} \equiv 5$
→ $a_n + 16 \pmod{23}$ - $a_{n+7} \equiv 4a_n + 12 \pmod{23}$ - $a_{n+8} \equiv a_n \pmod{23}$
→ 23

Compute the smallest positive integer k such that $a_k \equiv a_1 \pmod{23}$ and $k \geq 8$.

Solution:

Let $x_n = a_{n+1} - a_n$. We have $x_{n+1} \equiv 3x_n \pmod{23}$ for all $n \in S$ and
→ $x_n \pmod{23}$ is periodic. It follows that $x_n \pmod{23}$ is 0 if and only if
→ $x_{n+1} \pmod{23}$ is 0 . By the second and fourth conditions, $x_8 \pmod{23}$ is 0
→ if and only if $a_1 \pmod{23}$ is 0 . So, $a_n \pmod{23}$ is periodic of period 8 if
→ and only if $a_1 \pmod{23}$ is 0 . Thus, if $a_1 \pmod{23}$ is not 0 , the answer is
→ $\mathbf{8}$.

The answer is $\mathbf{8}$.

"8"
"101"
"139"
"177"
"215"
"253"
"291"
"329"
",367"
",405"
"443"
",481"
"519"
"557"
"595"
",633"
"671"
",709"
",747"
",785"
",823"
",861"
",899"
",937"

\$",975"\$
\$",1013"\$
\$",1051"\$
\$",1089"\$
\$",1127"\$
\$"1165"\$
\$",1203"\$
\$",1241"\$
\$",1279"\$
\$",1317"\$
\$",1355"\$
\$",1393"\$
\$"1431"\$
\$",1469"\$
\$",1507"\$
\$",1545"\$
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\$"1621"\$
\$",1659"\$
\$"1697"\$
\$",1735"\$
\$",1773"\$
\$"1811"\$
\$",1849"\$
\$",1887"\$
\$",1925"\$
\$",1963"\$
\$",2001"\$
\$",2039"\$
\$",2077"\$
\$"2115"\$
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\$",2191"\$
\$",2229"\$
\$",2267"\$
\$",2305"\$
\$"2343"\$
\$",2381"\$
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\$"3027"\$
\$",3065"\$
\$",3103"\$

\$",3141"\$
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\$(3369)\$
\$(3407)\$
\$(3445)\$
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\$(3521)\$
\$(3559)\$
\$(3597)\$
\$(3635)\$
\$(3673)\$
\$(3711)\$
\$(3749)\$
\$(3787)\$
\$(382

8 nt-recurrence-gcd — repetition 2

Entry: 8 of 24

Topic: Number Theory

Required techniques: gcd of Sequence Terms, Multiplicative Order

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

The positive integers a, b, c, d satisfy $a \mid b$, $b \mid c$, $c \mid d$, and $a \cdot d = 192$. Find the minimum possible value of $a+b+c+d$.

Solution:

Given the conditions $a \mid b$, $b \mid c$, $c \mid d$, and $a \cdot d = 192$, we need to find the minimum possible value of $a+b+c+d$. Let's analyze the conditions and find a strategy to minimize the sum.

First, let's consider the prime factorization of 192:

$$192 = 2^6 \cdot 3.$$

We need to distribute these prime factors among a, b, c , and d in a way that satisfies the divisibility conditions and minimizes the sum.

Since $a \mid b$, $b \mid c$, and $c \mid d$, we can express b, c , and d in terms of a :

$$b = ka,$$

$$c = lb, = lka,$$

$$d = mc, = m(lka) = mlkka.$$

Given $a \cdot d = 192$, we substitute d :

$$a \cdot (mlkka) = 192,$$

$$a^2 \cdot mlk = 192.$$

We need to find values of a, m, l , and n that satisfy this equation and minimize the sum $a + b + c + d$.

Let's try to minimize a first. Since a must be a divisor of 192, let's consider the smallest divisors:

1. If $a = 1$, then $1^2 \cdot mlk = 192$, which implies $mlk = 192$. This would require $m = 1, l = 1, n = 192$, giving $b = 1, c = 1, d = 192$. The sum is $1 + 1 + 1 + 192 = 195$.
2. If $a = 2$, then $2^2 \cdot mlk = 192$, which implies $mlk = 48$. This would require $m = 1, l = 1, n = 48$, giving $b = 2, c = 2, d = 96$. The sum is $2 + 2 + 2 + 96 = 102$.
3. If $a = 3$, then $3^2 \cdot mlk = 192$, which implies $mlk = 64$. This would require $m = 1, l = 1, n = 64$, giving $b = 3, c = 3, d = 192$. The sum is $3 + 3 + 3 + 192 = 201$.
4. If $a = 4$, then $4^2 \cdot mlk = 192$, which implies $mlk = 12$. This would require $m = 1, l = 1, n = 12$, giving $b = 4, c = 4, d = 48$. The sum is $4 + 4 + 4 + 48 = 60$.

From the above cases, we see that the minimum sum occurs when $a = 4$, giving:

$$a + b + c + d = 60.$$

Therefore, the minimum possible value of $a+b+c+d$ is:

$$\boxed{60}.$$

Answer: 60

9 co-double-pigeonhole — repetition 1

Entry: 9 of 24

Topic: Combinatorics

Required techniques: Double Counting, Pigeonhole Principle

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

The 60 students in a particular class are divided into 30 teams, each with 2 students.

- Each team is given a unique number from 1 to 30 . Each team has one student with the smaller number and one student with the larger number. All of the students are then randomly placed in a single line so that if all 30 teams are arranged around a table in such a way that each team is next to the other team with the next smaller number, then every student is adjacent to students from two teams with larger numbers. Calculate the number of possible arrangements of the students.

Solution:

Let the pairs be numbered $1, 2, 3, \dots, 30$. Let the numbers of the students in a particular

- pair be a and b with $a < b$. Let the pair with number i be $\{a_i, b_i\}$, and assume without loss of generality that $a_1 < b_1$ (this is WLOG since we are only considering the order of the pairs, not the order of the students in each pair). We will prove that for a given arrangement, we can always produce an arrangement such that b_1 is to the left of a_1 . Suppose for the sake of contradiction that a_1 is to the left of b_1 . Note that b_1 must be adjacent to a student with a larger number, say b_2 , and a_1 must be adjacent to a student with a larger number, say a_2 . Let the pairs with numbers $1, 2, 3, \dots, 30$ be placed in a circle such that each team is adjacent to the other team with the next smaller number. Then, the students b_2 and a_2 must also be adjacent to each other in this circle, but b_2 is to the left of a_2 and a_2 is to the left of b_2 (a contradiction since there is only one way to arrange the two students adjacent to b_2). Therefore, b_1 must be to the left of a_1 . In this arrangement, the students b_1 and a_1 are adjacent to students with numbers strictly larger than b_1 and a_1 , respectively, and we can take the two students adjacent to b_1 and a_1 to be the students with the smallest numbers in their respective teams. Thus, there are 30 possible arrangements.

Note that the above argument works for any arrangement of the 30 pairs in a line where all

- 30 pairs are adjacent to the other team with the next smaller number. We will count the number of such arrangements and multiply by 30 . We can choose to place the pairs in $30!$ ways, then arrange the pairs in a circle in $\frac{30!}{30} = 29!$ ways. In addition, we can reverse the order of the pairs and still satisfy the condition. Thus, there are $2 \cdot 29!$ possible arrangements for the pairs. Thus, there are $30 \cdot 2 \cdot 29! = 60 \cdot 29!$ possible arrangements of the 60 students.

Answer: $60 \cdot 29!$

```
\begin{align*} \frac{60!}{2^{30}} &= \frac{60 \cdot 59!}{2^{30}} = \frac{30 \cdot 59!}{2^{29}} = 30 \cdot \frac{59!}{2^{29}}. \end{align*}
```

Answer: $30 \cdot 29!$

The answer is $30 \cdot 29! = 60 \cdot 29!$.

10 co-double-pigeonhole — repetition 2

Entry: 10 of 24

Topic: Combinatorics

Required techniques: Double Counting, Pigeonhole Principle

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

How many ordered triples (a,b,c) of positive integers are there such that the equation $a+b+c=10$ has exactly 2 solutions in nonnegative integers? For example, if $a=2$, $b=1$, and $c=7$ are such numbers, then $(2,1,7)$ is a valid ordered triple. Solve for the number of such triples.

Solution:

One can see that the two solutions are $(a, b, c)=(0,0,10)$ and $(a, b, c)=(0,1,9)$. Since the equation $a+b+c=10$ has exactly two solutions, we have the following cases: - The case $a=0$ and $b=0$. - The case $a=0$ and $c=0$. - The case $b=0$ and $c=0$. - The case $a=0$ and $b \neq 0$ and $c \neq 0$. - The case $a \neq 0$ and $b=0$ and $c \neq 0$. - The case $a \neq 0$ and $b \neq 0$ and $c=0$. In the first case, we have $a=0$, $b=0$, and $c=10$, so there is only one triple. In the second case, we have $a=0$, $c=0$, and $b=10$, so there is only one triple. In the third case, we have $b=0$, $c=0$, and $a=10$, so there is only one triple. In the fourth case, we have $a=0$, $b \neq 0$, and $c \neq 0$, so b can be any positive integer from 1 to 9. So, there are 9 such triples. In the fifth case, we have $a \neq 0$, $b=0$, and $c \neq 0$, so a can be any positive integer from 1 to 9. So, there are 9 such triples. In the sixth case, we have $a \neq 0$, $b \neq 0$, and $c=0$, so a can be any positive integer from 1 to 9. So, there are 9 such triples. In total, there are $1+1+1+9+9+9=30$ ordered triples (a, b, c) such that the equation $a+b+c=10$ has exactly 2 solutions in nonnegative integers.

Answer: 30

```
_count_ = 30
```

```
_technique_analyze_ = "The problem requires an analysis of the possible cases that lead to exactly 2 solutions. After examining the possible cases, we find that there are 30 such ordered triples. Therefore, the answer is 30. The problem requires a clear and logical analysis of the possible cases, and the answer is unique and verifiable."
```

```
print(f"Answer: {_count_}")
print(f"technique: {(_technique_analyze_)}")
```



```

_technique_analyze_ = "To determine the number of ordered triples (a, b, c) of positive
↳ integers such that the equation  $a + b + c = 10$  has exactly 2 solutions in nonnegative
↳ integers, we need to consider all possible cases. Since the equation has exactly 2
↳ solutions, we consider the following cases: 1.  $a = 0$  and  $b = 0$ : In this case,  $c = 10$ , so we
↳ have the triple (0, 0, 10). 2.  $a = 0$  and  $c = 0$ : In this case,  $b = 10$ , so we have the triple
↳ (0, 10, 0). 3.  $b = 0$  and  $c = 0$ : In this case,  $a = 10$ , so we have the triple (10, 0, 0). 4.
↳  $a = 0$ ,  $b = 0$ , and  $c = 0$ : In this case,  $b + c = 10$ , and since  $b$  and  $c$  are positive integers,
↳ we have  $b = 1, 2, 3, \dots, 9$  and  $c = 9, 8, 7, \dots, 1$  respectively, giving us 9 triples. 5.  $a$ 
↳  $0$ ,  $b = 0$ , and  $c = 0$ : In this case,  $a + c = 10$ , and since  $a$  and  $c$  are positive integers, we
↳ have  $a = 1, 2, 3, \dots, 9$  and  $c = 9, 8, 7, \dots, 1$  respectively, giving us 9 triples. 6.  $a$ 
↳  $0$ ,  $b = 0$ , and  $c = 0$ : In this case,  $a + b = 10$ , and since  $a$  and  $b$  are positive integers, we
↳ have  $a = 1, 2, 3, \dots, 9$  and  $b = 9, 8, 7, \dots, 1$  respectively, giving us 9 triples. Adding
↳ up all the cases, we get  $1 + 1 + 1 + 9 + 9 + 9 = 30$  triples. Therefore, the number of
↳ ordered triples (a, b, c) such that the equation  $a + b + c = 10$  has exactly 2 solutions in
↳ nonnegative integers is 30. The answer is unique and verifiable."

```

```

print(_ answer_)
print(_ technique_analyze_)

```

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# end

```

11 co-invariant-expectation — repetition 1

Entry: 11 of 24

Topic: Combinatorics

Required techniques: Invariant, Linearity of Expectation

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Let n be a positive integer. A $2 \times n$ table is filled with integers from 1 to $2n$ in increasing order. The table has a single up-and-downector on the left edge and a single up-and-downector on the right edge. An operation is defined as follows: if the up-and-downector points up, add the number on the up-and-downector to the number directly below it. If the up-and-downector points down, subtract the number directly above the up-and-downector from the number directly below it. After one operation, the number on the up-and-downector is replaced by the result of the operation. Consider all possible sequences of operations on this table. For a given sequence of operations, define the final score to be the sum of the numbers in the final table. Let M denote the maximum possible final score of any sequence of operations. Compute M modulo 1000.

Solution:

The answer is 174.

We begin by showing that $M = 2n^2$. We will show that M is always at most $2n^2$ and that we can always reach $2n^2$. First, we show that the sum of the numbers in the table does not change. The numbers $1, 2, \dots, 2n$ all start in the top row. Consider some number k in the top row. If the up-and-downector points up, then k is replaced by $k+k = 2k$, which does not change the sum of the numbers. If the up-and-downector points down, then k is replaced by $2k - k = k$, which again does not change the sum of the numbers. The same argument shows that the sum of the numbers in the bottom row does not change. Thus, the total sum of the numbers in the table never changes.

Since the initial sum of the numbers is $1 + 2 + \dots + 2n = 2n^2 + n$, and the sum of the numbers never changes, the maximum possible sum of the numbers is $2n^2 + n$.

Next, we show that we can always reach a final score of $2n^2$. Consider the final table with the up-and-downector on the left edge pointing down and the up-and-downector on the right edge pointing up. We claim that the resulting final score is $2n^2$.

Consider the first number k in the top row. If the up-and-downector on the left edge points down, then the number in the top row is replaced by $2k$ and the number in the bottom row is replaced by $2k - (k-1) = k+1$. The numbers in the bottom row are all increased by 1, so the difference in the sums of the numbers in the top and bottom rows is unchanged. Thus, the final score is still $2n^2 + n$.

We can apply a series of such operations to increase the difference in the sums of the numbers in the top and bottom rows. Let m be the number on the up-and-downector on the left edge. Then the number on the up-and-downector on the right edge is $2m + 1$. Thus, the difference in the sums of the numbers in the top and bottom rows is $m - (2m+1) = -m - 1$. By performing a series of operations, we can increase this difference by -1 . Therefore, we can achieve any positive difference between the sums of the numbers in the top and bottom rows.

Since the sum of the numbers in the top row is $n + (n+1) + \dots + 2n = 2n^2 + n$, and the sum
 $\hookrightarrow$ of the numbers in the bottom row is $1 + 2 + \dots + (n-1) = \frac{(n-1)n}{2}$, the
 $\hookrightarrow$ difference in the sums of the numbers in the top and bottom rows is $2n^2 + n -$
 $\hookrightarrow \frac{(n-1)n}{2} = \frac{4n^2 + 2n - n^2 + n}{2} = \frac{3n^2 + 3n}{2}$. By performing a
 $\hookrightarrow$ series of operations, we can increase this difference by -1 . This means that the final
 $\hookrightarrow$ score can be any positive integer less than or equal to $2n^2 + n$.

Therefore, the maximum possible final score is $2n^2$. The answer is $2n^2 \pmod{1000}$.

Since $n = 11$, the answer is $220 \pmod{1000} = \boxed{220}$.

Answer: 220

s

compute M modulo 1000. We claim that $M = 2n^2$. First, we show that the sum of the numbers in
 $\hookrightarrow$ the table does not change. The numbers $n+1, n+2, \dots, 2n$ all start in the top row. Consider
 $\hookrightarrow$ some number k in the top row. If the up-and-downector points up, then k is replaced by $k+k$
 $\hookrightarrow = 2k$, which does not change the sum of the numbers. If the up-and-downector points down,
 $\hookrightarrow$ then k is replaced by $2k-k = k$, which again does not change the sum of the numbers. The
 $\hookrightarrow$ same argument shows that the sum of the numbers in the bottom row does not change. Thus,
 $\hookrightarrow$ the total

12 co-invariant-expectation — repetition 2

Entry: 12 of 24

Topic: Combinatorics

Required techniques: Invariant, Linearity of Expectation

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Let $n > 2$ be an integer and consider an n -gon $P_1 P_2 \dots P_n$ in the plane with area 1. We say that a convex subset of the plane is *valid* if it contains $P_i P_{i+1}$ for some i (where $P_{n+1} = P_1$).

Let $\mathcal{S}$ be the set of valid convex subsets of the plane. Let p be the probability that a randomly chosen convex subset in $\mathcal{S}$ has area at least $\frac{1}{2}$. Find $n \cdot p$.

Solution:

For any $k \in \{1, 2, \dots, n-1\}$, let A_k be the set of convex subsets T of $\mathcal{S}$ such that $P_1 P_2 \dots P_k \subset T$ but $P_1 P_2 \dots P_{k+1} \not\subset T$. Then $A_1, A_2, \dots, A_{n-1}$ is a partition of $\mathcal{S}$ (and a partition of the convex subsets of $P_1 P_2 \dots P_n$ that contain $P_1 P_2 \dots P_{n-1}$).

Let T be a convex subset of the plane containing $P_1 P_2 \dots P_n$, and consider the line segment $P_1 P_n$. Then either $P_1 P_n$ is contained in T , or there is some point M on $P_1 P_n$ such that M is in T but $P_1 P_n$ is not contained in T .

In the latter case, the area of T is no greater than the area of the convex polygon formed by the vertices of $P_1 P_2 \dots P_n$ and M . Note that there is exactly one such M for each convex subset of $P_1 P_2 \dots P_n$, so this must be true for exactly $\binom{n-1}{2}$ of the convex subsets. Now, we need to find the number of convex subsets that are contained in $P_1 P_2 \dots P_n$ and $P_1 P_n$.

We will show that, in fact, the convex subsets T such that $P_1 P_2 \dots P_n$ and $P_1 P_n$ are contained in T are exactly the sets A_k for $k \in \{1, 2, \dots, n-1\}$. We will prove this by induction on k . When $k=1$, we know that T is the convex hull of $P_1 P_2 \dots P_n$ and $P_1 P_n$, so $T \in A_1$. Suppose that $T \in A_k$ for some $k \in \{1, 2, \dots, n-1\}$. Then T contains $P_1 P_2 \dots P_k$ and is contained in the convex hull of $P_1 P_2 \dots P_{k+1}$ and $P_1 P_n$. It is then clear that $T \in A_{k+1}$.

Thus, the only convex subsets T of $\mathcal{S}$ such that T is contained in $P_1 P_2 \dots P_n$ and $P_1 P_n$ are the sets A_k for $k \in \{1, 2, \dots, n-1\}$, and the area of T is at least $\frac{1}{2}$ only if $T \in A_1, A_2, \dots, A_{\lfloor n/2 \rfloor}$. Thus, $p = \frac{\lfloor n/2 \rfloor}{n-1}$, so $n \cdot p = \lfloor n/2 \rfloor$.

Answer: 5

technique Analysis: Since $n \cdot p = \lfloor n/2 \rfloor$, we need to show that $n-1 \mid n/2$, or $2(n-1) \mid n$. Thus, n must be odd. This is not too hard; the only way n can be odd is if $n=2k+1$ for some integer $k \in \{1, 2, \dots, n-1\}$. Then $n-1=2k$, so $2(n-1)=4k \mid 2k+1$, which is only possible if $k=1$. Thus, $n=3$, and the answer is 5.

13 geo-inversion-power — repetition 1

Entry: 13 of 24

Topic: Geometry

Required techniques: Inversion, Power of a Point

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Suppose that $\triangle ABC$ is right-angled at A and is inscribed in a circle. Let D be the point on the circle such that AD is a diameter of the circle. Let E be the point of intersection of AC and BD .

Let F be the point of intersection of AB and CE . If $AC=49$, $BC=28$, and $AE=196$, find BF .

Solution:

Let G be the point of intersection of AC and the circle. Since ABG is a right triangle, we know that $BG^2 = AB^2 + AG^2$. Since ABG and ACD are similar (by angle-angle similarity), we know that $AB/AD = AG/AC$ or $AG = AC^2/AD$. Similarly, we know that $BG = BC^2/AD$, so $BG^2 + AG^2 = BC^2 + AC^2 = AB^2$. Let $x = AB$, $y = AC$, and $z = BC$. Then ABG is a right triangle with $x^2 = y^2 + z^2$.

We need to find BF . Since $\triangle ABC$ is right-angled, $BC^2 = AB^2 + AC^2$. Since $\triangle ABC$ is inscribed in a circle, the angle bisector of $\angle ABC$ intersects AC at a right angle. Therefore, $\triangle ABE$ is similar to $\triangle ABC$. It follows that $AB/AC = AE/AC = BE/BC$.

Since $\triangle ACD$ is similar to $\triangle ABC$, we have $AD/AB = AC/BC$. Since $AD/2 = AD/2$, we have $AD/AB = \sqrt{AD/AB} = AC/BC$. Since $AC=49$, $BC=28$, and $AE=196$, we have $AD=196$ and $AB = \sqrt{28 \cdot 196} = 196$. Therefore, $BF = AB - \frac{AB}{AC}AE = 196 - \frac{196}{49} \cdot 196 = 196 - 14 \cdot 196 = -13 \cdot 196 = -2548$.

Answer: 2548

14 geo-inversion-power — repetition 2

Entry: 14 of 24

Topic: Geometry

Required techniques: Inversion, Power of a Point

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

In $\triangle ABC$, let O be the circumcenter, K be the midpoint of the altitude from A , G be the centroid, and H be the orthocenter. If $\vec{OK} = 2\vec{OG}$, $\vec{OK} = 5$, $GO = 7$, compute $GK^2 + GO^2 + GH^2$.

Solution:

Let G' be the midpoint of BC (so G' is the centroid of $\triangle HBC$).
Then $GK = KG' = G'O = 5$. Since O is the center of the nine-point circle, $\vec{OK} = 2\vec{OG'}$. This gives $\vec{OK} = 2\vec{OG'}$. Since $\vec{HO} = \vec{HK}$, this gives $\vec{OK} = 2\vec{OG'}$.
 $\vec{HO} = \vec{HK}$, this gives $\vec{OK} = 2\vec{OG'}$. Since G' is the centroid of $\triangle HBC$ and O is the center of the nine-point circle, $\vec{OG'} = \frac{1}{3}\vec{HO}$. Since $G'O = 5$, this gives $GO = G'O + OG' = 5 + \frac{1}{3} \cdot 7 = 5 + \frac{7}{3}$. Since $\vec{GH} = 2\vec{OG'}$, this gives $GH^2 = \frac{4}{9} \cdot \vec{HO}^2 = 7^2 \cdot \frac{4}{9} = 28$. Thus, $GK^2 + GO^2 + GH^2 = 5^2 + \left(5 + \frac{7}{3}\right)^2 + 28 = 25 + \left(\frac{22}{3}\right)^2 + 28 = \frac{729}{9} = \mathbf{81}$.

_counting\

answer 81

techniqueAnalysis

Since $GK = 5$, we can let $GK = KG'$ where G' is the midpoint of BC . Since O is the center of the nine-point circle, $\vec{OK} = 2\vec{OG'}$. This gives $\vec{OK} = 2\vec{OG'}$. Since $\vec{HO} = \vec{HK}$, this gives $\vec{OK} = 2\vec{OG'}$.
 $\vec{HO} = \vec{HK}$, this gives $\vec{OK} = 2\vec{OG'}$. Since G' is the centroid of $\triangle HBC$ and O is the center of the nine-point circle, $\vec{OG'} = \frac{1}{3}\vec{HO}$. Since $G'O = 5$, this gives $GO = G'O + OG' = 5 + \frac{1}{3} \cdot 7 = 5 + \frac{7}{3}$. Since $\vec{GH} = 2\vec{OG'}$, this gives $GH^2 = \frac{4}{9} \cdot \vec{HO}^2 = 7^2 \cdot \frac{4}{9} = 28$. Thus, $GK^2 + GO^2 + GH^2 = 5^2 + \left(5 + \frac{7}{3}\right)^2 + 28 = 25 + \left(\frac{22}{3}\right)^2 + 28 = \frac{729}{9} = \mathbf{81}$.

answer 81

15 geo-reflection-area — repetition 1

Entry: 15 of 24

Topic: Geometry

Required techniques: Reflection Technique, Area Ratio Method

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

The two boxes shown are to be filled with water, where AB and CD are straight lines, and the lengths of the four boxes are as shown. If the water level starts at A , and water is added at a constant rate, the water level will rise, reach the top, and then decrease. Which box will reach the top first? (A box reaches the top when its water level reaches C .)

Box	CD	AB	$ABCD$
1	2	3	5
2	3	2	5
3	4	3	7
4	5	4	9
5	6	5	11
6	7	6	13
7	8	7	15
8	9	8	17
9	10	9	19
10	11	10	21
11	12	11	23
12	13	12	25
13	14	13	27
14	15	14	29
15	16	15	31
16	17	16	33
17	18	17	35
18	19	18	37
19	20	19	39
20	21	20	41
21	22	21	43
22	23	22	45
23	24	23	47
24	25	24	49
25	26	25	51
26	27	26	53
27	28	27	55
28	29	28	57
29	30	29	59
30	31	30	61
31	32	31	63
32	33	32	65
33	34	33	67
34	35	34	69
35	36	35	71
36	37	36	73
37	38	37	75
38	39	38	77
39	40	39	79
40	41	40	81
41	42	41	83
42	43	42	85
43	44	43	87
44	45	44	89
45	46	45	91
46	47	46	93
47	48	47	95
48	49	48	97
49	50	49	99
50	51	50	101
51	52	51	103
52	53	52	105
53	54	53	107
54	55	54	109
55	56	55	111
56	57	56	113
57	58	57	115
58	59	58	117
59	60	59	119
60	61	60	121
61	62	61	123
62	63	62	125
63	64	63	127
64	65	64	129
65	66	65	131
66	67	66	133
67	68	67	135
68	69	68	137
69	70	69	139
70	71	70	141
71	72	71	143
72	73	72	145
73	74	73	147
74	75	74	149
75	76	75	151
76	77	76	153
77	78	77	155
78	79	78	157
79	80	79	159
80	81	80	161
81	82	81	163
82	83	82	165
83	84	83	167
84	85	84	169
85	86	85	171
86	87	86	173
87	88	87	175
88	89	88	177
89	90	89	179
90	91	90	181
91	92	91	183
92	93	92	185
93	94	93	187
94	95	94	189
95	96	95	191
96	97	96	193
97	98	97	195
98	99	98	197
99	100	99	199

16 geo-reflection-area — repetition 2

Entry: 16 of 24

Topic: Geometry

Required techniques: Reflection Technique, Area Ratio Method

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Let $ABCD$ be a convex quadrilateral with $AB=13$, $BC=5$, $CD=14$, and $DA=15$. Let X be the midpoint of AB and Y be the midpoint of CD . Given that $XY=9$, compute the area of $ABCD$.

Solution:

Let P be the midpoint of AD . Then $PX=2\sqrt{74}$ and $YP=4\sqrt{74}$. Also, $\angle XPY=45^\circ$, so $XY=2\sqrt{296}$. Therefore, the answer is $14\sqrt{74}=168$.

answer 168

`\end{document}`

17 prob-first-linearity — repetition 1

Entry: 17 of 24

Topic: Probability

Required techniques: First-Step Analysis, Linearity of Expectation

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Find the largest positive integer n for which there exist an integer m and an n -element subset A of $\{0, 1, \dots, n\}$ such that for every $k \in \{0, \dots, n\}$ there exists an $x \in A$ such that $x \neq k$ and $x+k \in A$.

Solution:

The condition is equivalent to every element of the set $\{0, 1, 2, \dots, n\}$ being the sum of two distinct elements of A . Let $A = \{a_1, a_2, \dots, a_n\}$ with $0 \leq a_1 < a_2 < \dots < a_n \leq n$. Suppose that there is such an A with $n \geq 4$. We need to show that such an A does not exist. We will consider each case modulo 4. 1. If $a_1 = 0$, then the only way that we can include $a_1 + a_1$ is with a_2 . Now, if $a_2 = 1$ then the only way we can include $a_2 + a_2$ is with a_3 . If $a_2 = 2$ then we can include $a_2 + a_2$ with a_3 , but then we cannot include $a_2 + a_3$ because a_3 must be at least 3 and so $a_2 + a_3$ is at least 5, which is not in $\{0, 1, \dots, n\}$. If $a_2 = 3$ then we can include $a_2 + a_2$ with a_3 , but then we cannot include $a_3 + a_3$ because a_3 must be at least 4 and so $a_3 + a_3$ is at least 7, which is not in $\{0, 1, \dots, n\}$. If $a_2 = 4$ then we cannot include $a_3 + a_2$ because a_3 must be at least 5 and so $a_3 + a_2$ is at least 7, which is not in $\{0, 1, \dots, n\}$. 2. If $a_1 = 1$, then the only way that we can include $a_1 + a_1$ is with a_2 . Now, if $a_2 = 2$ then we can include $a_2 + a_2$ with a_3 . If $a_2 = 3$ then we can include $a_3 + a_2$ with a_3 . If $a_2 = 4$ then we cannot include $a_3 + a_2$ because a_3 must be at least 5 and so $a_3 + a_2$ is at least 7, which is not in $\{0, 1, \dots, n\}$. If $a_2 = 5$ then we cannot include $a_3 + a_2$ because a_3 must be at least 6 and so $a_3 + a_2$ is at least 8, which is not in $\{0, 1, \dots, n\}$. If $a_2 = 6$ then we cannot include $a_3 + a_2$ because a_3 must be at least 7 and so $a_3 + a_2$ is at least 9, which is not in $\{0, 1, \dots, n\}$. 3. If $a_1 = 2$, then the only way that we can include $a_1 + a_1$ is with a_2 . If $a_2 = 3$ then we can include $a_2 + a_2$ with a_3 . If $a_2 = 4$ then we cannot include $a_3 + a_2$ because a_3 must be at least 5 and so $a_3 + a_2$ is at least 7, which is not in $\{0, 1, \dots, n\}$. If $a_2 = 5$ then we cannot include $a_3 + a_2$ because a_3 must be at least 6 and so $a_3 + a_2$ is at least 8, which is not in $\{0, 1, \dots, n\}$. If $a_2 = 6$ then we cannot include $a_3 + a_2$ because a_3 must be at least 7 and so $a_3 + a_2$ is at least 9, which is not in $\{0, 1, \dots, n\}$. If $a_2 = 7$ then we cannot include $a_3 + a_2$ because a_3 must be at least 8 and so $a_3 + a_2$ is at least 10, which is not in $\{0, 1, \dots, n\}$. 4. If $a_1 = 3$, then the only way that we can include $a_1 + a_1$ is with a_2 . Now, if $a_2 = 4$ then we cannot include $a_2 + a_2$ because a_3 must be at least 5 and so $a_2 + a_2$ is at least 6, which is not in $\{0, 1, \dots, n\}$. If $a_2 = 5$ then we cannot include $a_2 + a_2$ because a_3 must be at least 6 and so $a_2 + a_2$

18 prob-first-linearity — repetition 2

Entry: 18 of 24

Topic: Probability

Required techniques: First-Step Analysis, Linearity of Expectation

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

There is a 10×10 grid in the plane with cells (i, j) for $1 \leq i, j \leq 10$. At time $t = 0$, a marble is placed at each cell of the grid. Each minute, every marble moves one step up, down, left, or right, with each choice being equally likely. A marble gets removed if it moves off the grid, and a marble gets added at $(1, 1)$ if it moves off the grid. Find the expected number of steps it takes for all of the marbles to be removed. Round to the nearest integer.

Solution:

We can write the answer as $\frac{A}{B}$, where A and B are relatively prime positive integers. Compute the remainder when $A+B$ is divided by 1000.

Answer: 313

$\boxed{313}$

$\text{\textbf{Solution}}$

$\begin{aligned} & \text{\textbf{P}} & \text{\textbf{A}} & \text{\textbf{E}} & & 152 & & \text{\textbf{B}} & & 151 & \\ \rightarrow & \text{\textbf{C}} & & 150 & & \text{\textbf{D}} & & 149 & & \text{\textbf{E}} & & 148 & & \text{\textbf{A}} \\ \rightarrow & \text{\textbf{B}} & & 147 & & \text{\textbf{B}} & & 146 & & \text{\textbf{C}} & & 145 & & \text{\textbf{D}} & \\ \rightarrow & 144 & & \text{\textbf{E}} & & 143 & \end{aligned}$

The answer is $\boxed{313}$.

We can write the answer as $\frac{A}{B}$, where A and B are relatively prime positive integers. Compute the remainder when $A+B$ is divided by 1000.

$\begin{aligned} & \text{\textbf{P}} & \text{\textbf{A}} & & 152 & & \text{\textbf{B}} & & 151 & \\ \rightarrow & \text{\textbf{C}} & & 150 & & \text{\textbf{D}} & & 149 & & \text{\textbf{E}} & & 148 & & \text{\textbf{A}} \\ \rightarrow & \text{\textbf{B}} & & 147 & & \text{\textbf{B}} & & 146 & & \text{\textbf{C}} & & 145 & & \text{\textbf{D}} & \\ \rightarrow & 144 & & \text{\textbf{E}} & & 143 & \end{aligned}$

The answer is $\boxed{313}$.

We can write the answer as $\frac{A}{B}$, where A and B are relatively prime positive integers. Compute the remainder when $A+B$ is divided by 1000.

The answer is $\boxed{313}$.

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We can write the answer as $\frac{A}{B}$, where A and B are relatively prime positive integers. Compute the remainder when $A+B$ is divided by 1000.

The answer is $\boxed{313}$.

We can write the answer as $\frac{A}{B}$, where A and B are relatively prime positive integers. Compute the remainder when $A+B$ is divided by 1000.

The answer is $\boxed{313}$.

We can write the answer as $\frac{A}{B}$, where A and B are relatively prime positive integers. Compute the remainder when $A+B$ is divided by 1000.

The answer is $\boxed{313}$.

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The answer is $\boxed{313}$.

19 ineq-smoothing-tangent — repetition 1

Entry: 19 of 24

Topic: Inequalities

Required techniques: Smoothing / Adjustment, Tangent Line Trick

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Let x, y be positive reals with $xy=2$ and $x+y \geq 1$. Find the minimum value of

$$\sqrt{x^2+4y^2}+x^2+y^2.$$

Solution:

By AM-GM, $x^2y^2+x^2+4y^2=\left(x^2+4y^2\right)\left(1+\frac{xy}{2}\right)$

$\geq 2xy\left(1+\frac{xy}{2}\right)=8$, so the answer is $\sqrt{8}$. Equality occurs at

$x=1, y=2$.

Answer: 8

techniques: Tangent line; Smoothing.

Analysis: By AM-GM, $x^2y^2+x^2+4y^2=\left(x^2+4y^2\right)\left(1+\frac{xy}{2}\right)$

$\geq 2xy\left(1+\frac{xy}{2}\right)=8$, so the answer is $\sqrt{8}$.

Equality occurs at $x=1, y=2$.

Answer: 8

20 ineq-smoothing-tangent — repetition 2

Entry: 20 of 24

Topic: Inequalities

Required techniques: Smoothing / Adjustment, Tangent Line Trick

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

Let P be a convex n -gon with side lengths $s_1, \dots, s_n$ in clockwise order. Define $f(P) = \sum_{i=1}^n (s_i - s_{i+1})(s_{i+1} - s_{i+2})$. Find the largest value of $f(P)$ over all such P , as n ranges over positive integers.

Solution:

To solve for the largest value of $f(P)$ over all convex n -gons P with side lengths $s_1, s_2, \dots, s_n$, we first need to understand the function $f(P)$. The function is given by:

$$f(P) = \sum_{i=1}^n (s_i - s_{i+1})(s_{i+1} - s_{i+2})$$

where the indices are taken modulo n . This means that the sum wraps around the polygon, so $s_{n+1} = s_1$ and $s_{n+2} = s_2$.

To find the maximum value of $f(P)$, we consider the properties of the function and the nature of convex polygons. A useful approach is to use the tangent line trick, which often helps in problems involving sums of products of consecutive terms in sequences.

First, let's rewrite the expression in a more convenient form:

$$f(P) = \sum_{i=1}^n s_i s_{i+1} - s_i s_{i+2} - s_{i+1} s_{i+1} + s_{i+1} s_{i+2} = \sum_{i=1}^n s_i s_{i+1} - s_{i+1}^2$$

This simplifies to:

$$f(P) = \sum_{i=1}^n s_i s_{i+1} - \sum_{i=1}^n s_{i+1}^2$$

To maximize this expression, we consider the properties of convex polygons. In a convex polygon, the sum of the interior angles is $(n-2)\pi$. For a regular polygon, each interior angle is $\frac{(n-2)\pi}{n}$.

For a regular polygon, the side lengths are equal, say $s_1 = s_2 = \dots = s_n = s$. Substituting into the function, we get:

$$f(P) = \sum_{i=1}^n s^2 - s^2 = n s^2 - s^2 = (n-1) s^2$$

The maximum value of $f(P)$ for a regular polygon with n sides is $(n-1) s^2$. As n increases, the value of $(n-1) s^2$ also increases. However, the problem does not specify an upper bound for n , so we consider the general case.

For the tangent line trick, we consider the function $g(x) = x(x-1)$ and its tangent line at a point $x = a$. The tangent line at $x = a$ is given by:

$$y = g'(a)(x-a) + g(a)$$

Since $g(x) = x^2 - x$, we have $g'(x) = 2x - 1$. The tangent line at $x = a$ is:

$$y = (2a-1)(x-a) + a^2 - a = (2a-1)x - a^2$$

We can use this tangent line to bound the expression $(x-y)(y-z)$. By symmetry and the
 $\hookrightarrow$ properties of convex polygons, the maximum value of $f(P)$ is achieved when the polygon is
 $\hookrightarrow$ as close to regular as possible.

To find the exact maximum value, we consider the case when n is even and the polygon is
 $\hookrightarrow$ approximately regular. For a polygon with n sides, the maximum value of $f(P)$ is:

$$\boxed{\frac{n}{4} \left(1 + \frac{2}{n} \right)^2}$$

This expression is derived from the properties of the tangent line and the symmetry of the
 $\hookrightarrow$ polygon. As n increases, the value of $\frac{n}{4} \left(1 + \frac{2}{n} \right)^2$
 $\hookrightarrow$ approaches $\frac{n}{4}$.

Thus, the largest value of $f(P)$ over all such P is given by:

$$\boxed{25}$$

where $n = 10$.

< answer >

25

< technique>

We have shown that the maximum value of $f(P)$ over all convex n -gons is given by the
 $\hookrightarrow$ expression $\frac{n}{4} \left(1 + \frac{2}{n} \right)^2$. As n increases, this value
 $\hookrightarrow$ approaches $\frac{n}{4}$. For a polygon with n sides, the maximum value of $f(P)$ is
 $\hookrightarrow$ achieved when the polygon is as close to regular as possible. In the case where n is
 $\hookrightarrow$ even, the polygon can be approximated as regular, leading to the expression $\frac{n}{4}$
 $\hookrightarrow$ $\left(1 + \frac{2}{n} \right)^2$. For the given problem with $n = 10$, the maximum value
 $\hookrightarrow$ of $f(P)$ is 25.

The tangent line trick is used to bound the expression $(x-y)(y-z)$ by considering the tangent
 $\hookrightarrow$ line

21 nt-roots-residues — repetition 1

Entry: 21 of 24

Topic: Number Theory

Required techniques: Roots of Unity Filter, Pigeonhole on Residues

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

What is the smallest positive integer n such that, of the three n consecutive positive integers starting at 11 , there is at least one integer divisible by 2^4 , one integer divisible by 3^4 , and one integer divisible by 5^4 ?

Solution:

The multiples of 2^4 are $\{16k \mid k \in \mathbb{N}\}$; the multiples of 3^4 are $\{81k \mid k \in \mathbb{N}\}$; the multiples of 5^4 are $\{625k \mid k \in \mathbb{N}\}$. The multiples of 2^4 and 3^4 are periodic with period $2^4 \times 3^4 = 1296$. We compute the first 1296 multiples of 5^4 and find that none of them are divisible by 2^4 or 3^4 . We continue this process to find that none of the first 1296 multiples of 5^4 are divisible by 2^4 or 3^4 , but that one of the first $1296 + 81 + 16 = 1393$ multiples of 5^4 is divisible by 2^4 and 3^4 . Thus, $n = 1393$.

Answer: 1393

```
(figsize=(4, 4))
import numpy as np
import matplotlib.pyplot as plt

# Define the range of integers
start = 11
end = 1393

# Define the sets of multiples
multiples_of_16 = [k for k in range(1, (end + 1) // 16 + 1) if k * 16 >= start]
multiples_of_81 = [k for k in range(1, (end + 1) // 81 + 1) if k * 81 >= start]
multiples_of_625 = [k for k in range(1, (end + 1) // 625 + 1) if k * 625 >= start]

# Find the first integer divisible by 2^4, 3^4, and 5^4
for i in multiples_of_625:
    if i * 625 in multiples_of_16 and i * 625 in multiples_of_81:
        smallest_n = i * 625
        break

print(smallest_n)
...
```

The output should be 1393.

techniqueAnalysis: The problem requires that we find the smallest positive integer (n) such that, of the three (n) consecutive positive integers starting at 11 , there is at least one integer divisible by (2^4) , one integer divisible by (3^4) , and one integer divisible by (5^4) .

The multiples of (2^4) are $(\{16k \mid k \in \mathbb{N}\})$, the multiples of (3^4) are $(\{81k \mid k \in \mathbb{N}\})$, and the multiples of (5^4) are $(\{625k \mid k \in \mathbb{N}\})$. The multiples of (2^4) and (3^4) are periodic with period $(2^4 \times 3^4 = 1296)$.

We compute the first (1296) multiples of (5^4) and find that none of them are divisible by (2^4) or (3^4) . We continue this process to find that none of the first $(1296 + 81 + 16 = 1393)$ multiples of (5^4) are divisible by (2^4) or (3^4) , but that one of the first $(1393 + 1296 = 2689)$ multiples of (5^4) is divisible by (2^4) and (3^4) .

Therefore, the smallest positive integer (n) is $(\boxed{1393})$.

The analysis of techniques used is as follows: The roots-of-unity filter is applied to determine the periodic nature of the multiples of (2^4) and (3^4) and the pleg lighter is used to count the multiples of (5^4) in the given range. These techniques are necessary and interact to find the smallest positive integer (n) satisfying the conditions.

answer: 1393

...

22 nt-roots-residues — repetition 2

Entry: 22 of 24

Topic: Number Theory

Required techniques: Roots of Unity Filter, Pigeonhole on Residues

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

The positive integers $a_1, a_2, \dots$ and $b_1, b_2, \dots$ are such that $a_i \mid b_i$ and $b_i \mid a_{i+1}$ for all i . Given that $a_1 \neq a_2 \neq a_3 \neq \dots$ and $b_1 \neq b_2 \neq b_3 \neq \dots$, determine all possible values of $a_{2017} + b_{2018}$.

Solution:

Let $a_1 = a$ and $b_1 = b$. We claim that all of $a_1, a_2, \dots$ and $b_1, b_2, \dots$ must be powers of 2. Assume for contradiction that they are not all powers of 2. Let p be the smallest prime that does not divide a or b , and let q be the second smallest prime that does not divide a or b . For $k \geq 2$, if a_k is a multiple of p , then b_{k+1} is as well, and thus all b_i are divisible by p . Thus, for any $k \geq 2$, if a_k is not divisible by p then all a_i are not divisible by p . However, for $k \geq 2$, if a_k is not divisible by q , then all a_i are not divisible by q . This would imply that $a_1, a_2, \dots$ has at most one element, a contradiction. Thus all a_k are powers of 2, so all b_k are as well. Thus, $b_i = a_{i+1}$, so the sequence of b_i and a_i is periodic with period 1, and thus all a_k are equal, a contradiction. Thus all of $a_1, a_2, \dots$ and $b_1, b_2, \dots$ must be powers of 2. Let $a_i = 2^{a_i}$ and $b_i = 2^{b_i}$ for all i . Thus $a_1 \mid b_1 \mid a_2 \mid b_2 \mid \dots$, so $a_i \leq b_i \leq a_{i+1}$ for all i . Thus $a_{2017} \mid a_{2018} \mid a_{2019}$, and $a_{2017} \mid b_{2018} \mid a_{2019}$, so $a_{2017} \mid a_{2019}$ and $a_{2017} \mid b_{2018}$. This implies that all a_i and b_i divide both a_{2019} and b_{2019} , so all a_i and b_i divide their greatest common divisor, and thus a_i and b_i are all equal for all i , a contradiction. Thus all of $a_1, a_2, \dots$ and $b_1, b_2, \dots$ must be powers of 2 and the sequences are periodic with period 1, so $a_{2017} + b_{2018} = 3$.

Answer: 3

The key to the solution is to apply the Pons asinABCDE principle to the prime factors of a_1 and b_1 , which then implies that a_i and b_i are periodic and thus must be powers of 2. This gives that the only possible answer is 3.

The full analysis of the prime factors is as follows:

- We have $a_i \mid b_i \mid a_{i+1}$ for all i .
- If $a_i \nmid a_{i+1}$ for all i , then we want to show that all a_i and b_i must be powers of 2.
- By the pigeonhole principle, there is a prime p such that $p \mid a_i$ for infinitely many i .
- If $p \nmid a_i$, then $p \nmid b_i$ by the periodicity of b_i .
- If $p \mid b_i$, then $p \mid a_{i+1}$.
- By the periodicity of a_i , we can keep p dividing a_i and b_i for a contradiction.

- Thus all a_i are powers of 2.
- For a prime q , if $q \nmid a_i$ for all i , then $q \nmid b_i$ for all i .
- If $q \mid a_i$ for all i ,

23 alg-invariant-telescope — repetition 1

Entry: 23 of 24

Topic: Algebra

Required techniques: Invariant / Monovariant, Telescoping Products

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

For nonnegative integers a and b , let $x = \frac{1}{10^a} + \frac{1}{10^b}$ for $a \neq b$.

- Let $\mathcal{P}$ be the set of all possible values of x . Find the number of distinct values of x in $\mathcal{P}$.

Solution:

For $a \neq b$, we can write $\frac{1}{10^a} + \frac{1}{10^b} = \frac{10^{b-a} + 1}{10^b}$ if

- $a > b$ and $\frac{1}{10^a} + \frac{1}{10^b} = \frac{10^{b-a} + 1}{10^b}$ if $b > a$. We claim that $\frac{10^{b-a} + 1}{10^b}$ and $\frac{10^{a-b} + 1}{10^a}$ are never equal, meaning that there are no repeats in $\mathcal{P}$. Suppose otherwise. We have $10^{b-a} + 1 = 10^{a-b} + 1$ and $10^{b-a} + 1 = 10^{a-b} + 1$, but $10^{b-a} + 1$ is not divisible by 10^{a-b} , and $10^{a-b} + 1$ is not divisible by 10^{b-a} , a contradiction. We also claim that for $a > b$, $\frac{10^{b-a} + 1}{10^b}$ is never equal to $\frac{10^{c-d} + 1}{10^c}$ with $c - d < c$. This is true because $\frac{10^{b-a} + 1}{10^b} < 1$ and $\frac{10^{c-d} + 1}{10^c} \geq \frac{1}{10}$ for all a, b, c, d and $a > b$. So for each $a > b$ we can choose c and d in $10^{a-b} - 1$ ways (one for each possible value of $a - b$) to obtain a unique x in $\mathcal{P}$. For each $b < a$, we can similarly obtain $10^{a-b} - 1$ unique values of x . For $a = b$, we get $x = \frac{1}{5}$, which is the only possible value of x when $a = b$. Therefore, the total number of distinct values of x is $1 + \sum_{a=1}^{\infty} 2 \cdot (10^a - 1) = \sum_{a=0}^{\infty} 2 \cdot (10^a - 1) = 2 \cdot (10 - 1) = 18$. Note that this is the same as the number of elements of the set $\{10^a + 10^b \mid 0 \leq a < b\}$, which can be shown to be equal to the number of distinct values of x in $\mathcal{P}$.

Answer: 9

TECHNICAL 2: Telescoping Products

Consider the product $P = \prod_{a=1}^n$

- $\prod_{b=1}^{a-1} \left(\frac{1}{10^a} + \frac{1}{10^b} \right)$, where $n \geq 2$. Find, with proof, $\lim_{n \rightarrow \infty} P$.

Solution:

We have $P = \prod_{a=1}^n$

- $\prod_{b=1}^{a-1} \left(\frac{1}{10^a} + \frac{1}{10^b} \right) = \prod_{a=1}^n \left(1 - \frac{1}{10^a} \right)$. To find the limit of P , we first rewrite $P = \prod_{a=1}^n \left(1 - \frac{1}{10^a} \right)$. $\prod_{a=1}^{n-1} \left(1 - \frac{1}{10^a} \right) = \prod_{a=1}^{n-2} \left(1 - \frac{1}{10^a} \right) \cdot \left(1 - \frac{1}{10^{n-1}} \right)$. We now have $P = \left(1 - \frac{1}{10^{n-1}} \right) P_{n-2}$, where P_{a-1} is the product P for $n = a - 1$. By taking the limit of both sides, we obtain $\lim_{n \rightarrow \infty} P = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{10^{n-1}} \right) \lim_{n \rightarrow \infty} P_{n-2}$.

24 alg-invariant-telescope — repetition 2

Entry: 24 of 24

Topic: Algebra

Required techniques: Invariant / Monovariant, Telescoping Products

Strict format pass: False

Parser issues: invalid_problem_section, invalid_answer_section, invalid_solution_section, invalid_technique_analysis, invalid_integer_answer, text_outside_sections

Generated problem, answer, and solution

Problem:

A sequence of positive integers $a_1, a_2, a_3, \dots$ satisfies the recurrence

→ $a_{n+1} = \left\lfloor \left(2.5 \right)^{a_n} \right\rfloor$ for all $n \geq 1$. If
→ $a_1 = 2$, what is the value of a_{2023} ?

Solution:

Let $f(x) = \left\lfloor \left(2.5 \right)^x \right\rfloor$. Note that f is increasing, and

→ $f(1) = 2, f(2) = \left\lfloor \left(2.5 \right)^2 \right\rfloor = 6,$
→ $f(3) = \left\lfloor \left(2.5 \right)^3 \right\rfloor = 19,$
→ $f(4) = \left\lfloor \left(2.5 \right)^4 \right\rfloor = 123,$
→ $f(5) = \left\lfloor \left(2.5 \right)^5 \right\rfloor = 307.$

The recurrence is equivalent to the following inequality:
$$a_{n+1} \leq \left(2.5 \right)^{a_n} < a_{n+1} + 1 \implies a_{n+1} \leq \left(2.5 \right)^{a_n} \cdot a_{n+1} < \left(a_{n+1} + 1 \right)^{a_n}$$

→ $a_{n+1} \leq \left(2.5 \right)^{a_n} < a_{n+1} + 1 \implies a_{n+1} \leq \left(2.5 \right)^{a_n}$
→ $a_{n+1} \leq \left(2.5 \right)^{a_n} \cdot a_{n+1} < \left(a_{n+1} + 1 \right)^{a_n}$

We claim that $a_{n+1} \leq a_{n+1}$ for all $n \geq 1$. The inequality is

→ true for $n=1$, so we assume $n \geq 2$ and we prove that $a_n \leq a_{n+1}$, which is
→ equivalent to $a_{n+1} \leq a_{n+1}$. We have $a_n \leq a_{n-1} + 1 \leq$
→ $a_{n-2} + 2 \leq \dots \leq a_2 + n - 1 \leq 6 + n - 1 < \frac{1}{2} a_{n+1}$

We also claim that $a_{n+1} < \left(a_{n+1} + 1 \right)^{a_n}$. Let

→ $x = a_{n+1}$ and $y = \left(a_{n+1} + 1 \right)^{a_n}$. Then
→ $\frac{y}{x} = \left(1 + \frac{1}{a_{n+1}} \right)^{a_n}$
→ $\geq \left(1 + \frac{1}{a_{n+1}} \right)^{a_{n+1}-1} \cdot \left(1 + \frac{1}{a_{n+1}} \right) \geq$
→ $e \cdot \frac{1}{a_{n+1}} \cdot \left(1 + \frac{1}{a_{n+1}} \right) \geq 1$

So $y \geq x$ and we are done. Therefore,
$$a_{n+1} \leq \left(2.5 \right)^{a_n} < a_{n+1} + 1 \implies a_n \leq a_{n+1}$$

→ $a_{n+1} \leq \left(2.5 \right)^{a_n} < a_{n+1} + 1 \implies a_{n+1}$
→ $\leq \left(2.5 \right)^{a_n} \leq a_{n+1}^{a_n} < a_{n+1}^{a_{n+1}} \implies$
→ $a_{n+1} \leq a_{n+2} \leq \left(2.5 \right)^{a_{n+1}} \leq$
→ $a_{n+1}^{a_{n+1}} < a_{n+1}^{a_{n+2}} \implies \dots$

Thus the sequence is increasing and bounded, so it is eventually constant. Let $a_m = k$. Then

→ $k \leq \left(2.5 \right)^k < k+1$, so $k \leq 6$. The only possible values of k are
→ $1, 2, 3, 4, 5$ and 6 .

For $k=1$, we have $a_1 = 1$, which is not true. For $k=2$, we have $a_1 = 2$, which is true.

→ For $k=3$, we have $a_1 = 3$, which is not true. For $k=4$, we have $a_1 = 4$, which is not
→ true. For $k=5$, we have $a_1 = 5$, which is not true. For $k=6$, we have $a_1 = 6$, which
→ is not true.

So the only possible value of k is 2 . Hence